Duality for transmuted preorder M-triangles

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Abstract

We prove Conjecture 8.2 of Athanasiadis and Chapoton for every finite preorder. The proof realizes a rank-and-support enumerator in a two-variable lattice-point series. The interior points of the associated cone are a fixed translate of all its lattice points. Stanley reciprocity then gives exactly the proposed transmuted M-triangle duality.

Task 1. Full solution

Let τ be a preorder on an n-element set E. Put

$$P_\tau = \{a \in \mathbb{Z}_{\geq 0}^E : a(I) \leq |I| \text{ for every order ideal } I \text{ of } \tau\}, \qquad a(I) = \sum_{e \in I} a_e.$$

Its order is componentwise and its rank is $|a| = a(E)$. Write $s(a) = |\operatorname{supp}(a)|$. For a finite graded poset P, define

$$M_P(x,y) = \sum_{a \leq b} \mu_P(a,b) x^{|a|} y^{|b|}, \qquad \overline{M}_P(x,y) = M_P\left(\frac{1-y}{1-xy}, 1-xy\right).$$

Theorem 1. *For every finite preorder τ of size n,*

$$\boxed{(xy)^n \overline{M}_{P_\tau}(1/x, 1/y) = \overline{M}_{P_\tau}(y,x).}$$

No lattice realization of P_τ is assumed.

We may suppose $n \geq 1$, since the empty preorder gives both sides equal to one. Introduce the homogeneous degree-n polynomial

$$H_\tau(X,Y,Z) = \sum_{a \in P_\tau} X^{n-|a|} Y^{s(a)} Z^{|a|-s(a)}. \tag{1}$$

All exponents are nonnegative, since $|a| \leq n$. Downward closure makes each interval $[a,b]$ a product of chains. Its Möbius function is zero unless every coordinate of $b-a$ is zero or one, in which case it is $(-1)^{|b|-|a|}$. Summing first over $a \leq b$ gives

$$M_{P_\tau}(x,y) = \sum_{b \in P_\tau} (xy)^{|b|} (1-x^{-1})^{s(b)}.$$

After transmutation this becomes the polynomial identity

$$\overline{M}_{P_\tau}(x,y) = \sum_{b \in P_\tau} (-y)^{s(b)} (1-x)^{s(b)} (1-y)^{|b|-s(b)} = H_\tau(1, y(x-1), 1-y). \tag{2}$$

We will prove the stronger homogeneous identity

$$H_\tau(X,Y,Z) = H_\tau(X+Y-Z, Z, Y). \tag{3}$$

A cone and its interior translation

Set $\sigma = \tau^*$, the reversed preorder. In coordinates $(t, \alpha, \beta) \in \mathbb{R} \times \mathbb{R}^E \times \mathbb{R}^E$ consider the rational cone

$$C = \{(t, \alpha, \beta) : \alpha, \beta \geq 0, \quad \beta(I) + \alpha(E \setminus I) \leq t \text{ for every ideal } I \text{ of } \sigma\}. \tag{4}$$

Here the empty ideal is included. The cases $I = \varnothing, E$ give $|\alpha|, |\beta| \leq t$. Thus C is pointed. It is full dimensional, since

$$w = (n+1, \mathbf{1}, \mathbf{1})$$

strictly satisfies all its inequalities. Every displayed normal is nonzero, so strict inequalities characterize the interior, even if some inequalities are redundant. For integer points, they are $\alpha_e, \beta_e \geq 1$ and $\beta(I) + \alpha(E \setminus I) \leq t - 1$. Subtracting w therefore gives the exact equality

$$C^\circ \cap \mathbb{Z}^{2n+1} = w + (C \cap \mathbb{Z}^{2n+1}). \tag{5}$$

Indeed, the left side of each ideal inequality drops by n, while the coordinate t drops by $n+1$.

Let $\alpha_0 = t - |\alpha|$. It is a nonnegative integer on lattice points of C. Define the two-variable series

$$G(u, v) = \sum_{(t,\alpha,\beta) \in C \cap \mathbb{Z}^{2n+1}} u^{\alpha_0} v^{|\alpha|}.$$

Every coefficient is finite: fixing α_0 and $|\alpha|$ fixes t, and both nonnegative vectors have sum at most t. Moreover a nonzero cone vector has positive total degree in this grading. Stanley's reciprocity theorem for rational cones [2, Theorem 1], followed by this monomial specialization, applies without a zero-degree ray. Since $\dim C = 2n+1$ and translation by w changes $(\alpha_0, |\alpha|)$ by $(1, n)$, (5) gives, as rational functions,

$$G(u^{-1}, v^{-1}) = -uv^n G(u, v). \tag{6}$$

The numerator of the series

For fixed nonnegative integers (α_0, α), the possible β form the lattice points of

$$\alpha_0 \Delta_E + \sum_{e \in E} \alpha_e \Delta_{U_\sigma(e)}, \qquad \Delta_A = \operatorname{conv}(0, \mathbf{e}_f : f \in A). \tag{7}$$

To check this, the cone inequalities become $\beta(I) \leq \alpha_0 + \alpha(I)$ for nonempty ideals. The transportation criterion for the Minkowski sum is $\beta(A) \leq \alpha_0 + \alpha(D_\sigma(A))$ for every nonempty subset A. These conditions are equivalent: take ideals for one direction and use $A \subseteq D_\sigma(A)$ for the other.

Postnikov's established formula in [1, Theorem 2.1 and Lemma 4.1], with these arbitrary summand weights, says that the number of points in (7) is

$$\sum_{a \in P_\tau} \binom{\alpha_0 + n - |a|}{n - |a|} \prod_{e \in E} \binom{\alpha_e + a_e - 1}{a_e}. \tag{8}$$

The additional draconian coordinate equals $n - |a|$; its summand is the universal simplex, which explains the first binomial factor. For fixed nonnegative k, the needed series are

$$\sum_{j \geq 0} \binom{j+k}{k} u^j = (1-u)^{-k-1}, \qquad \sum_{j \geq 0} \binom{j+k-1}{k} v^j = \begin{cases} (1-v)^{-1} & k = 0, \\ v(1-v)^{-k-1} & k > 0. \end{cases}$$

In the $k = 0$ case the binomial coefficient is one, including $j = 0$. Sum (8) with weights $u^{\alpha_0} v^{|\alpha|}$. For $r = |a|$ and $s = s(a)$, its contribution is $(1-u)^{-n+r-1} v^s (1-v)^{-n-r}$. Consequently

$$G(u,v) = \frac{H_\tau(1-v, v(1-u), 1-u)}{(1-u)^{n+1}(1-v)^{2n}}. \tag{9}$$

The root-polytope interpretation and its one-variable support enumerator appear in [3, Section 4.2]; here we retain the extra grading needed for the M-triangle.

Reciprocity becomes transmuted duality

Homogeneity of H_τ and (9) give

$$G(u^{-1}, v^{-1}) = -uv^n \frac{H_\tau(u(1-v), 1-u, v(1-u))}{(1-u)^{n+1}(1-v)^{2n}}.$$

Cancel the common factors using (6). We obtain

$$H_\tau(1-v, v(1-u), 1-u) = H_\tau(u(1-v), 1-u, v(1-u)). \tag{10}$$

The first coordinate on the right equals $(1-v) + v(1-u) - (1-u)$, so this is (3) on the displayed parametrization. Homogeneity extends it to all triples. Explicitly, for generic (X, Y, Z) take

$$v = Y/Z, \qquad u = 1 - (Z-Y)/X, \qquad \lambda = XZ/(Z-Y).$$

Then $(X, Y, Z) = \lambda(1-v, v(1-u), 1-u)$. Thus the two polynomials in (3) agree on a dense open set and agree identically.

Finally, substitute $(X, Y, Z) = (1, x(y-1), 1-x)$ into (3), and use (2) and homogeneity:

$$\begin{aligned} \overline{M}_{P_\tau}(y,x) &= H_\tau(1, x(y-1), 1-x) \\ &= H_\tau(xy, 1-x, x(y-1)) \\ &= (xy)^n H_\tau\left(1, \frac{1-x}{xy}, \frac{y-1}{y}\right) = (xy)^n \overline{M}_{P_\tau}(1/x, 1/y). \end{aligned}$$

This proves Theorem 1 as a polynomial identity.

Verification and scope

The exact verifier checks 78,480 Möbius intervals and all 389 labeled preorders of sizes 1–4. It checks 2,466 weighted cone sections containing 74,376 lattice points, including their interior translations; the source's five-element triangle and larger examples also agree. All 26 pages of the primary and all 25 pages of the later root-polytope paper were read. The latter proves support-enumerator duality, a different statement. Bounded checks found no earlier full target claim or target journal final. Review is by the authoring agent only.

Tasks 2–5. Proposed follow-up directions

Task 2. Describe the coefficient relations in (3) directly in terms of ranks, support sizes, and slack coordinates.

Task 3. Retain separate weights on the supply coordinates and derive a multivariate refinement of the transmuted duality.

Task 4. Characterize bipartite graphs whose augmented root cone admits an interior translation with the same grading shift.

Task 5. Determine which of these reciprocal triangles are realized by finite graded lattices, addressing the additional existence question.

References

[1] C. A. Athanasiadis and F. Chapoton, *Polytopes and posets associated to preorders*, arXiv:2605.26916v1, 2026.

[2] M. Beck and M. Develin, *On Stanley's reciprocity theorem for rational cones*, arXiv:math/0409562v3, 2005.

[3] Z. Dai, Q. Hou, Z. Liu, W. Thawinrak, and H. Wang, *Counting Lattice Points in Minkowski Sums of Cross Polytopes*, arXiv:2608.16037v2, 2026.